

become more effective and are positive and thus PV increases with increase in P after reaching a minimum. This suggests the existence of intermolecular attraction because below T_B , the gases are highly compressible.

2. **Above Boyle Temperature ($T > T_B$):** Above Boyle temperature, PV continuously increases with increase in P . This is because, the coefficient B is +ve above Boyle temperature and thus the terms on R.H.S. of equation (2.1) are positive. It now indicates that the intermolecular attraction is now less significant.

3. **At Boyle temperature ($T = T_B$):** The product PV is almost constant for a long range of pressure. Thus, Boyle's law is obeyed over a wide range of pressures.

Thus, below the Boyle temperature, the gases are highly compressible and this suggests the existence of intermolecular attraction. Beyond Boyle temperature, Boyle's law is obeyed and intermolecular attractions are less significant.

Definition: Boyle temperature is that temperature above which a real gas behaves like an ideal gas and obeys Boyle's law.

2.7 Reasons for Modification of Gas Equation

The perfect gas equation $PV = RT$ is not obeyed by real gases, particularly at high pressure and low temperature. At ordinary temperatures and pressures, the deviation is small but at high pressures and low temperature, the deviations are very large. In the derivation of the perfect gas equation, on the basis of kinetic theory of gases, the following assumptions have been made, which do not hold true in case of real gases. These are (i) the size of the molecule of the gas is negligible, and (ii) the forces of intermolecular attraction are absent. But in actual practice, at high pressure the size of the molecule of the gas becomes significant and cannot be neglected, in comparison with the volume of the gas. Also at high pressure, the molecules come closer and the forces of molecular attraction are appreciable. Hence, a perfect gas equation needs modification.

2.8 Van der Waals Equation of State

In 1879 Van der Waals modified the perfect gas equation $PV = RT$ by applying corrections for

1. Intermolecular force of attraction and
2. Finite size of molecules.

1. Correction for Pressure:

Consider a molecule A of the gas, well inside the vessel as shown in Fig. 2.7. It is attracted by other molecules in all directions with the same force and the net force acting on it is zero. But when it strikes the wall of the vessel (position A'), it is pulled back by other molecules. Its velocity and hence momentum with which it will strike the wall would be less than the momentum with which it will strike in the absence of the force of attraction. This reduction in momentum results in decrease of pressure. In other words, the observed pressure of the gas is less than the actual pressure. The correction for pressure p depends upon

(i) No. of molecules striking a unit area per unit time of the walls of the container, and

(ii) The resultant inward pull of cohesion on each of the striking molecules.

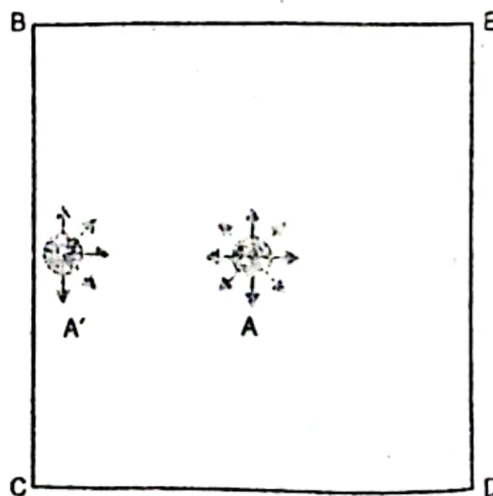


Fig. 2.7

Each of these factors is proportional to the density of the gas.

$$\therefore \text{Correction for pressure } p \propto \rho^2 \propto \frac{1}{V^2}$$

$$\therefore p = \frac{a}{V^2}$$

Here a is a constant and V is the volume of the gas.

Therefore, the corrected or real pressure

$$= P + p = \left(P + \frac{a}{V^2} \right) \quad \dots(2.3)$$

where P is the observed pressure.

2. Correction for Volume :

Due to the finite size of gas molecules, the actual space available for the movement of the molecules is less than the volume of the vessel. The molecules have the sphere of influence (Fig. 2.8) around them of radius ($2r$), within which no other molecule can penetrate.

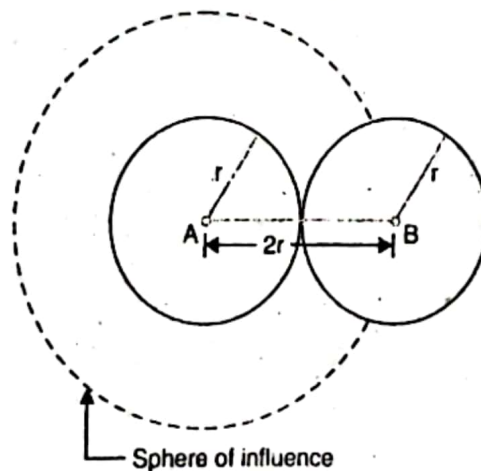


Fig. 2.8

Here r is the radius of each gas molecule.

$$\text{Volume of the molecule} = x = \frac{4}{3} \pi r^3$$

The centre of any two molecules can approach each other only by a minimum distance of $2r$. The volume of sphere of influence of each molecule

$$s = \frac{4}{3} \pi (2r)^3 = 8 \times \frac{4}{3} \pi r^3 = 8x$$

Let us fill the whole space of the volume V with n molecules one by one.

The volume available for the first molecule = V

The volume available for the second molecule = $V - 8x$
 $= V - s$

The volume available for n th molecule = $[V - (n-1)s]$.

$\therefore$ Average space available for each molecule

$$= \frac{V + (V - s) + (V - 2s) + \dots + [V - (n-1)s]}{n}$$

$$= \frac{nV}{n} - \frac{s}{n} (1 + 2 + 3 + \dots + (n-1))$$

$$\begin{aligned}
 &= V - \frac{s}{n} \cdot \frac{(n-1)n}{2} \\
 &= V - \frac{ns}{2} + \frac{s}{2}
 \end{aligned}$$

As the number of molecules is very large $\frac{s}{2}$ can be neglected.

∴ Average space available of each molecule

$$\begin{aligned}
 &= V - \frac{ns}{2} \\
 &= V - \frac{n(8x)}{2} \quad (\because s = 8x) \\
 &= V - 4(n x) \\
 &= V - b \quad \dots(2.4)
 \end{aligned}$$

Where $b = 4(n x) =$ four times the actual volume of the molecules.

Thus, the Van der Waals equation of state for a gas is

$$\left(P + \frac{a}{V^2}\right)(V - b) = RT \quad \dots(2.5)$$

where a and b are Van der Waals constants.

2.9 Comparison with Experimental PV Curves

The validity of Van der Waals equation of state for a gas is tested by comparing its PV curves with the experimental results.

The Van der Waals equation of state is given by

$$\left(P + \frac{a}{V^2}\right)(V - b) = RT$$

or

$$P = \frac{RT}{(V - b)} - \frac{a}{V^2} \quad \dots(2.6)$$

we see that if $V \rightarrow \infty, P \rightarrow 0$. Again if V is very small and $V \rightarrow b$, then $P \rightarrow \infty$.

A family of plots i.e., curves between pressure and volume at various temperatures are drawn using equation 2.5. The curves are as shown in Fig. 2.9. In the graph, the horizontal portion is absent. But in place of it, the curve $ABCDE$ is obtained. This does not agree with the experimental isothermals for CO_2 as obtained by Andrew. However, the portion AB has been explained as due to super cooling of the vapours and the portion ED is due to super-heating of the liquid. But the portion BCD cannot be explained because it shows decrease in volume with decrease in pressure, which is not true in actual practice. At higher temperatures, however, both theoretical and practical PV curves are similar.

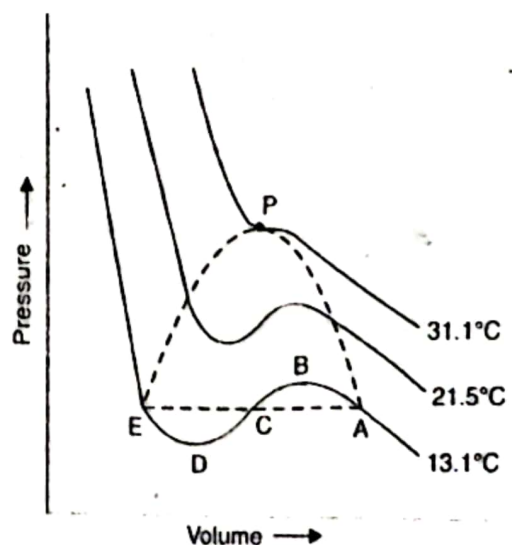


Fig. 2.9

2.10 Estimation of Critical Constants

Consider the critical isothermal ACB as shown in Fig. 2.10. At the critical point C, the curve is horizontal. Therefore, at point C, $\left(\frac{dP}{dV}\right) = 0$. At this point, the tangent also crosses the curve. Therefore, the tangent at such a point is said to be stationary and the point is called 'the point of inflexion'. At the point of inflexion, we have

$$\frac{d^2P}{dV^2} = 0$$

Hence, at critical point C, we have $\frac{dP}{dV} = 0$ and $\frac{d^2P}{dV^2} = 0$.

The Van der Waals equation is

$$P = \frac{RT}{(V-b)} - \frac{a}{V^2} \quad \dots(i)$$

$$\therefore \frac{dP}{dV} = -\frac{RT}{(V-b)^2} + \frac{2a}{V^3} \quad \dots(ii)$$

$$\text{and} \quad \frac{d^2P}{dV^2} = +\frac{2RT}{(V-b)^3} - \frac{6a}{V^4} \quad \dots(iii)$$

At the critical point C, we have

$$T = T_c, \quad P = P_c, \quad V = V_c, \quad \frac{dP}{dV} = 0 \text{ and } \frac{d^2P}{dV^2} = 0$$

Substituting in the above equations,

$$P_c = \frac{RT_c}{(V_c-b)} - \frac{a}{V_c^2} \quad \dots(iv)$$

$$\frac{RT_c}{(V_c-b)^2} = \frac{2a}{V_c^3} \quad \dots(v)$$

$$\text{and} \quad \frac{2RT_c}{(V_c-b)^3} = \frac{6a}{V_c^4} \quad \dots(vi)$$

Dividing (v) by (vi),

$$\frac{V_c-b}{2} = \frac{V_c}{3} \quad \text{or} \quad 3V_c - 3b = 2V_c$$

$$\therefore \quad V_c = 3b \quad \dots(2.7)$$

Substituting the value of V_c in equation (v), we get

$$\frac{RT_c}{(3b-b)^2} = \frac{2a}{(3b)^3}$$

$$\frac{RT_c}{4b^2} = \frac{2a}{27b^3}$$

$$\therefore \quad T_c = \frac{8a}{27bR} \quad \dots(2.8)$$

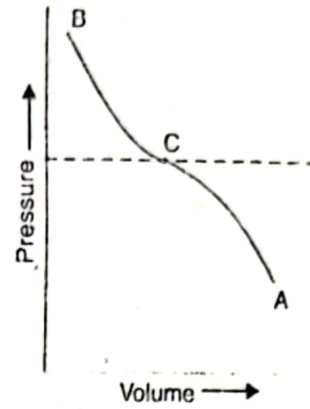


Fig. 2.10

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Substituting the value of V_c and T_c in equation (iv), we get

$$\begin{aligned} P_c &= \frac{8aR}{27bR(3b-b)} - \frac{a}{(3b)^2} \\ &= \frac{8a}{27 \times 2b^2} - \frac{a}{9b^2} \\ &= \frac{a}{9b^2} \left[\frac{8}{6} - 1 \right] = \frac{a}{9b^2} \times \frac{1}{3} \end{aligned}$$

$$P_c = \frac{a}{27b^2}$$



...(2.9)

TABLE 2.1 Critical Temperature and Pressure of Common Gases

Substance	Critical Temperature °C	Critical Pressure (atmosphere)
Helium	- 268°C	2.26
Hydrogen	- 240°C	12.80
Nitrogen	- 146°C	33.50
Air	- 140°C	39.00
Argon	- 122°C	48.00
Oxygen	- 118°C	50.00
Carbon dioxide	31°C	73.00
Ammonia	130°C	15.00
Chlorine	146°C	76.00
Water	374°C	218.00

2.11 Constants of Van der Waals Equation

From equation (2.8),

$$T_c = \frac{8a}{27bR}$$

$$\therefore \frac{a}{b} = \frac{27RT_c}{8} \quad \text{...(vii)}$$

But

$$P_c = \frac{a}{27b^2}$$

$$\therefore \frac{a}{b^2} = 27 P_c \quad \text{...(viii)}$$

Dividing (vii) by (viii),

$$\frac{a}{b} \times \frac{b^2}{a} = \frac{27RT_c}{8} \times \frac{1}{27P_c}$$

$$b = \frac{RT_c}{8P_c}$$

... (2.10)



Fig. 2.11. Vander Waals coefficients

 Substituting for b in equation (vii), we get

$$\frac{a}{RT_c} = \frac{27 RT_c}{8P_c}$$

$$a = \frac{RT_c}{8P_c} \times \frac{27 RT_c}{8}$$

$$a = \frac{27 R^2 T_c^2}{64 P_c}$$

... (2.11)

 Equation (2.10) and (2.11) gives the value of constants a and b respectively.

2.12 Critical Coefficient

The quantity $\frac{RT_c}{P_c V_c}$ is called the critical coefficient. Its value is obtained by substituting the values of P_c , V_c and T_c .

$$\begin{aligned} \therefore \text{Critical coefficient} &= \frac{RT_c}{P_c V_c} = \frac{R \times \frac{8a}{27bR}}{\frac{a}{27b^2} \times 3b} \\ &= \frac{8a}{27b} \times \frac{27b^2}{3ab} = \frac{8}{3} \\ &= 2.667 \end{aligned}$$

... (2.12)

The calculated value of critical coefficient $\frac{RT_c}{P_c V_c}$ is $\frac{8}{3}$ and it is the same for all gases.

The experimental values of critical coefficient for different gases are given in Table 2.2.

TABLE 2.2

Experimental values of Critical Coefficient $\frac{RT_c}{P_c V_c}$

Substance	T_c in K	P_c in atm. in cm ³ /g	Specific Volume	$\frac{RT_c}{P_c V_c}$
Helium	5.1	2.25	15.4	3.13
Hydrogen	33.1	12.8	32.2	3.28
Nitrogen	125.9	33.5	3.21	3.42
Oxygen	154.2	49.7	2.32	3.42
Carbon dioxide	304	72.8	2.17	3.48
Water	647	218	3.181	4.30

NOTE: Here V_c = molecular wt. $\times$ specific volume.

The experimental value of the critical coefficient of all gases is greater than the theoretical value of 2.667.

2.13 Limitations of Van der Waals Equation

1. The values of constants 'a' and 'b' obtained by different methods differ considerably. Although Van der Waals theory assumes them to be constant. The value of a is found to depend on the temperature. At very high temperatures, it tends to zero.

2. According to theory, $V_c = 3b$, but it is found to depend on the nature of gas. Experimentally, it is found that $V_c \approx 2b$.

3. The theoretical value of critical coefficient $\frac{RT_c}{P_c V_c}$ is 2.667 for all gases. However, it varies from gas to gas with an average value of 3.7 for most of the gases. This constant appears to depend on the molecular structure of the gas.

Thus no gas obeys Van der Waals equation closely in the vicinity of the critical point.

2.14 Reduced Equation of State

When the actual pressure, volume and temperature of a fixed mass of a gas are divided by the critical values of these quantities, the resulting ratios are called the reduced pressure (P_r), reduced volume (V_r) and reduced temperature (T_r) respectively.

Thus, we have

$$P_r = \frac{P}{P_c}, \quad V_r = \frac{V}{V_c} \quad \text{and} \quad T_r = \frac{T}{T_c}$$

$$\therefore P = P_c P_r = \frac{a}{27b^2} P_r$$

$$V = V_c V_r = 3b V_r$$

$$T = T_c T_r = \frac{8a}{27Rb} T_r$$

Thus, the force is proportional to $\frac{1}{r^7}$.

3. Interaction between non-polar molecules

The time average electric moments of non-polar molecules are zero. Since the electrons are in the rotational motion around the nuclei even in the lowest electronic state, this motion gives rise to momentary electric dipole moment in a non-polar molecule. A momentary dipole moment of one molecule produces an electric field at another molecule. Consequently the electric dipoles give rise to the so called *dispersion forces*. By the procedure similar to that used in the previous case, it can be shown that this force is also proportional to the seventh power of the distance of separation between two molecules.

NUMERICAL EXAMPLES

EXAMPLE 2.1

Calculate the Van der Waals constants for dry air, given that $T_c = 132$ K, $P_c = 37.2$ atmospheres and R per mole = $82.07 \text{ cm}^3 \text{ atmos K}^{-1}$. (Nagpur 2005, Kanpur 1997)

SOLUTION. Given:

$$P_c = 37.2 \text{ atmospheres}$$

$$T_c = 132 \text{ K}$$

$$R = 82.07 \text{ cm}^3 \text{ atmos K}^{-1}$$

(i)

$$a = \frac{27}{64} \times \frac{R^2 T_c^2}{P_c}$$

$$= \left(\frac{27}{64} \right) \frac{(82.07)^2 (132)^2}{37.2}$$

$$a = 13.31 \times 10^6 \text{ atmos cm}^6$$

(ii)

$$b = \frac{RT_c}{8P_c}$$

$$= \frac{82.07 \times 132}{8 \times 37.2}$$

$$b = 36.41 \text{ cm}^3$$

EXAMPLE 2.2

Calculate critical temperature for CO_2 , given that $a = 0.00874 \text{ atmos, cm}^6$ and $b = 0.0023 \text{ cm}^3$. (Lucknow 1998)

SOLUTION. Critical temperature is given by

$$T_c = \frac{8a}{27Rb}$$

To evaluate the value of gram molecular gas constant R at a pressure of 1 atmosphere for a gram molecular volume at 0°C or 273 K from Van der Waals equation

$$\left(P + \frac{a}{V^2} \right) (V - b) = RT$$

Substituting the values,

$$\left(1 + \frac{0.00874}{1} \right) (1 - 0.0023) = R \times 273$$

$\therefore$

$$R = \frac{1.00874 \times 0.9977}{273}$$

Thus $T_c = \frac{8a}{27Rb} = \frac{8 \times 0.0874 \times 273}{27 \times 1.00874 \times 0.9977 \times 0.023}$
 $= 305.7 \text{ K}$
 $\therefore T_c = 32.7^\circ\text{C}$

EXAMPLE 2.3

The Van der Waals constants a and b for 1 gram molecule of hydrogen are $a = 0.245 \text{ atm-litre}^2 \cdot \text{mole}^{-2}$ and $b = 2.67 \times 10^{-2} \text{ litre-mole}^{-1}$. Calculate the critical temperature.

(Kerala 2001)

SOLUTION. Critical temperature

$$T_c = \frac{8a}{27Rb}$$

Here,

$$a = 0.245 \times 76 \times 13.6 \times 980 \times 10^6 \text{ dynes-cm}^4 \cdot \text{mole}^{-2}$$

$$b = 2.67 \times 10^{-2} \times 10^3 \text{ cm}^3 \cdot \text{mole}^{-1}$$

$$R = 8.31 \times 10^7 \text{ ergs/mole-K}$$

$$\therefore T_c = \frac{8 \times 0.245 \times 76 \times 13.6 \times 980 \times 10^6}{27 \times 8.31 \times 10^7 \times 2.67 \times 10^{-2} \times 10^3}$$

$$T_c = 33.14 \text{ K} = -239.86^\circ\text{C}$$

$$8 \times 0.245$$

$$27 \times 8.31 \times 10^7 \times 2.67 \times 10^{-2} \times 10^3$$

EXAMPLE 2.4

Van der Waals' constants of a gas are $a = 0.245 \text{ liter}^2 \text{ atm} \cdot \text{mole}^{-2}$, $b = 2.67 \times 10^{-10} \text{ litre} \cdot \text{mole}^{-1}$. The initial volume and pressure of the gas are 2 litre and 4 atmosphere respectively. The gas is made to suffer a free expansion to tripple its volume. Find the change in temperature of the gas. Given $C_v = 5 \text{ cal} \cdot \text{deg}^{-1} \cdot \text{mole}^{-1}$.

(Rohilkhand 1996)

SOLUTION. The Joule's coefficient for a real gas is given by

$$\eta = \left(\frac{\partial T}{\partial V} \right)_U = -\frac{1}{C_v} \cdot \frac{a}{V^2}$$

or

$$dT = -\frac{1}{C_v} \cdot \frac{a}{V^2} dV$$

Here,

$$a = 0.245 \text{ litre}^2 \cdot \text{atm} \cdot \text{mole}^{-2}$$

$$= 0.245 \times (10^3)^2 \times 10^{-12} \times 10^5 \text{ N m}^4 \cdot \text{mole}^{-2}$$

$$= 0.245 \times 10^{-1} \text{ Nm}^4 \cdot \text{mole}^{-2}$$

$$= 0.0245 \text{ Nm}^4 \cdot \text{mole}^{-2}$$

$$C_v = 5 \text{ cal} \cdot \text{deg}^{-1} \cdot \text{mole}^{-1}$$

$$V = 2 \text{ litre} = 2 \times 10^3 \text{ cm}^3 = 2 \times 10^{-3} \text{ m}^3$$

$$dV = 2 \times 3 - 2 = 4 \text{ litre} = 4 \times 10^{-3} \text{ cm}^3 = 4 \times 10^{-3} \text{ m}^3$$

Substituting, the change in temperature will be

$$dT = -\frac{1}{5 \times 4.2} \times \frac{0.0245}{(2 \times 10^{-3})^2} \times 4 \times 10^{-3}$$

$$= -\frac{0.0245 \times 4 \times 10^{-3}}{5 \times 4.2 \times 4}$$

$$= -1.17 \text{ K} = -1.17^\circ\text{C}$$

Thus, the temperature will fall by 1.17°C .

EXAMPLE 2.5

Van der Waal's constants for a gas are $a = 6.9 \times 10^{-2} \text{ J m}^3 \text{ mole}^{-2}$ and $b = 2.9 \times 10^{-5} \text{ m}^3 \text{ mole}^{-1}$. The universal gas constant $R = 8.31 \text{ J mole}^{-1} \text{ K}^{-1}$. Calculate the critical temperature of the gas. (Nagpur 2004, Kerala 2001)

SOLUTION. Critical temperature

$$\begin{aligned} T_c &= \frac{8a}{27Rb} \\ &= \frac{8 \times 6.9 \times 10^{-2} \text{ J m}^3 \text{ mole}^{-2}}{27 \times 8.31 \text{ J mole}^{-1} \text{ K}^{-1} \times 2.9 \times 10^{-5} \text{ m}^3 \text{ mole}^{-1}} \\ &= 84.83 \text{ K} \end{aligned}$$

EXAMPLE 2.6

Calculate the values of the constants a and b in Van der Waals' equation for dry air, given that for dry air, $T_c = 132 \text{ K}$, $P_c = 37.2 \text{ atm}$, R per mole = $82.07 \text{ cm}^3 \text{ atoms K}^{-1}$.

(Nagpur 2005/winter)

SOLUTION. (i)

$$\begin{aligned} a &= \frac{27 R^2 T_c^2}{64 P_c} \\ &= \frac{27 \times (82.07)^2 \times (132)^2}{64 \times 37.2} \\ &= 1.33 \times 10^6 \text{ Amts cm}^6/\text{mole} \end{aligned}$$

(ii)

$$\begin{aligned} b &= \frac{RT_c}{8P_c} \\ &= \frac{37.2 \times 132}{8 \times 37.2} \\ &= 36.40 \text{ c.c./mole} \end{aligned}$$

EXERCISE-II

- 2.1 In what way a real gas differs from an ideal gas. (Meerut 2000, 1999, G.N.D.U. 2000)
- 2.2 Describe Andrew's experiments on carbon dioxide. Discuss the results obtained. Show that the liquid and the gaseous states are two distant stages of a continuous phenomenon. (Agra 1995, Kerala 2001, Meerut 2000, Bangalore 2000)
- 2.3 Differentiate between gas and vapour using Andrew's experimental isothermals for CO_2 .
- 2.4 What are critical constants of a gas? (Alagappa 2004)
Define : (i) Critical temperature T_c
(ii) Critical pressure P_c
(iii) Critical volume V_c
- 2.5 Define Boyle temperature and explain the behavior of real gases above and below the Boyle temperature. (Calcutta 2004)
- 2.6 Is the Boyle temperature different for different gases? Explain why.
- 2.7 Explain the reasons for modification of perfect gas equation. (Nagpur 2005, 2004)
- 2.8 Derive and discuss the Van der Waals equation of state of a gas. Compare the results with Andrew's experimental PV curves. (Patna 2005, Nagpur 2005, 2004)
- 2.9 What do you understand by Virial coefficients? What is the value of first Virial coefficient? How does the second Virial coefficient vary with temperature? (Calcutta 2001)

2.10 Express the Van der Waals' equation in the following Virial form:

$$PV = R + \left[1 + \frac{B}{V} + \frac{C}{V^2} + \dots \right]$$

where $B, C, \dots$ are Virial coefficients. From above equation find out the expression of the second virial coefficient B and the Boyle temperature. What is the significance of Boyle temperature?

(Calcutta 2004)

2.11 Distinguish between a perfect gas and a real gas. Derive Van der Waals' equation of state and use it to obtain the expressions for the critical constants in terms of the constants of the Van der Waals' equation.

(Calcutta 2005, Ranchi 2004, M.D.U. 2003, 2001, 2000, 1999, Kerala 2001, K.U. 2001, Utkal 2003, Nagpur 2003, 2001)

2.12 What are the critical constants of a gas? Calculate the values of these constants in terms of the constants of the Van der Waals equation.

(Nagpur 2005, Alagappa 2004, Calcutta 2003)

2.13 Define critical coefficient of a gas. Is it same for all gases? Does experimental values agree with the theoretical value?

(Nagpur 2005)

2.14 Mention limitations of Van der Waals equation.

(Nagpur 2004, Rahilkhand 1998, Garhwal 2001)

2.15 Derive Van der Waals reduced equation of state of a gas.

(Kerala 2001, Garhwal 2000)

2.16 What is Joule-Thomson effect? How is it experimentally established? How will you interpret the effect?

(Nagpur 2004, Bangalore 2005, Mysore 2005, Patna 2005, Madras 2004)

2.17 Describe the porous plug experiment. What conclusions have been drawn from it? What is inversion temperature?

(Bangalore 2004, 2005, Alagappa 2004, Ranchi 2004, Madras 2005)

2.18 Define Joule-Thomson coefficient.

(Garhwal 2005, Nagpur 2004)

2.19 Give the theory of Joule-Thomson effect. Hence explain why hydrogen and helium show heating effect at ordinary temperatures.

(Nagpur 2004, Garhwal 2000, 2002, 2005, Bangalore 2002)

2.20 Give the theory of the porous plug experiment. Obtain conditions for heating and cooling of a gas.

(Gujrat 2005, Alagappa 2005, Bangalore 2002, 2004)

2.21 What are the critical constants of a gas? State and explain Van der Waals equation. Calculate the critical constants of gas in terms of the constants of this equation.

2.22 What is Joule-Thomson effect? Give the theory of Joule-Thomson effect. Hence explain the concept temperature of inversion.

(Garhwal 1996, Meerut 1996, Kanpur 1997, Kuvempa Uni 2005)

2.23 How is the ideal gas equation modified when mutual attraction between molecules and the finite size of the molecules are taken into consideration? Obtain relations between critical constants and the constants appearing in the Van der Waals equation.

(Kanpur 1997, 1999)

2.24 Derive the reduced equation of state for a gas starting from Van der Waals equation of state. Show that if the two gases have the same reduced pressure and volume, they also have the same reduced temperature?

(Kerala 2001, Garhwal 2000)

2.25 Starting from elementary ideas, set up Van der Waals equation of state and show how the constants of the equation can be expressed in terms of the critical constants. What is the experimental bearing of this equation?

(Garhwal 1995, Kanpur 1996, Meerut 2000)

2.26 Explain the corrections introduced by Van der Waals in the gas equation. Show that for a gas obeying Van der Waals equation

$$\frac{RT_c}{P_c V_c} = \frac{8}{3}$$

(Utkal 2003, K.U. 2001, Nagpur 2003, 2004, Meerut 2002, 2000, Kerala 2001, Calcutta 2003)

2.27 Draw Andrew's curves for CO_2 at different temperatures and obtain expression for critical constants from them. Show that

$$\frac{T_i}{T_c} = \frac{27}{4}$$

(Meerut 1999)

2.28 Derive and discuss the Van der Waals' equation of state of a gas. Mention its defects.

(Calcutta 2003)